

INTRODUCTION

Multiple sclerosis (MS) is an autoimmune disease affecting the central nervous system and **more than 2.8 million people worldwide** (He et al., 2026). MS has a strong genetic component, with numerous immune-related genes linked to disease susceptibility. However, **most research has focused on individual risk genes** rather than how combinations of genes distinguish MS patients from healthy controls.

IDENTIFYING EFFECTIVE MULTI-GENE SIGNATURES COULD IMPROVE THE MOLECULAR CLASSIFICATION OF MS

LITERATURE REVIEW

MS has a strong genetic and immune component (Sawcer et al., 2011). Four immune-related genes have been repeatedly associated with MS:

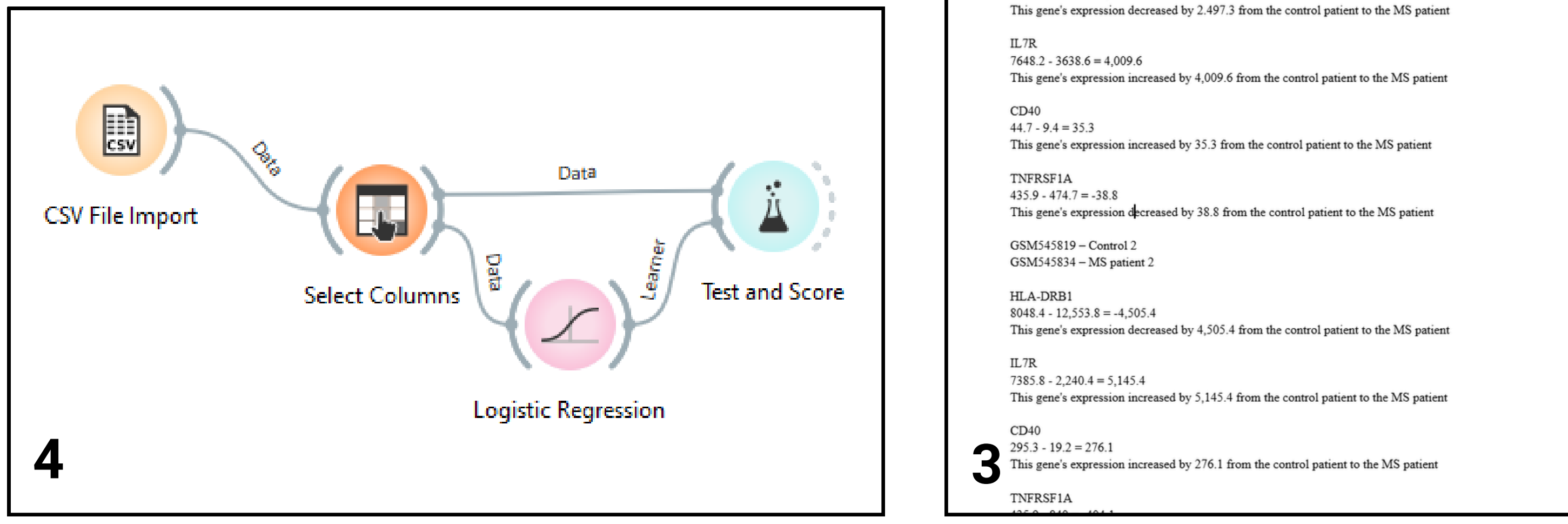
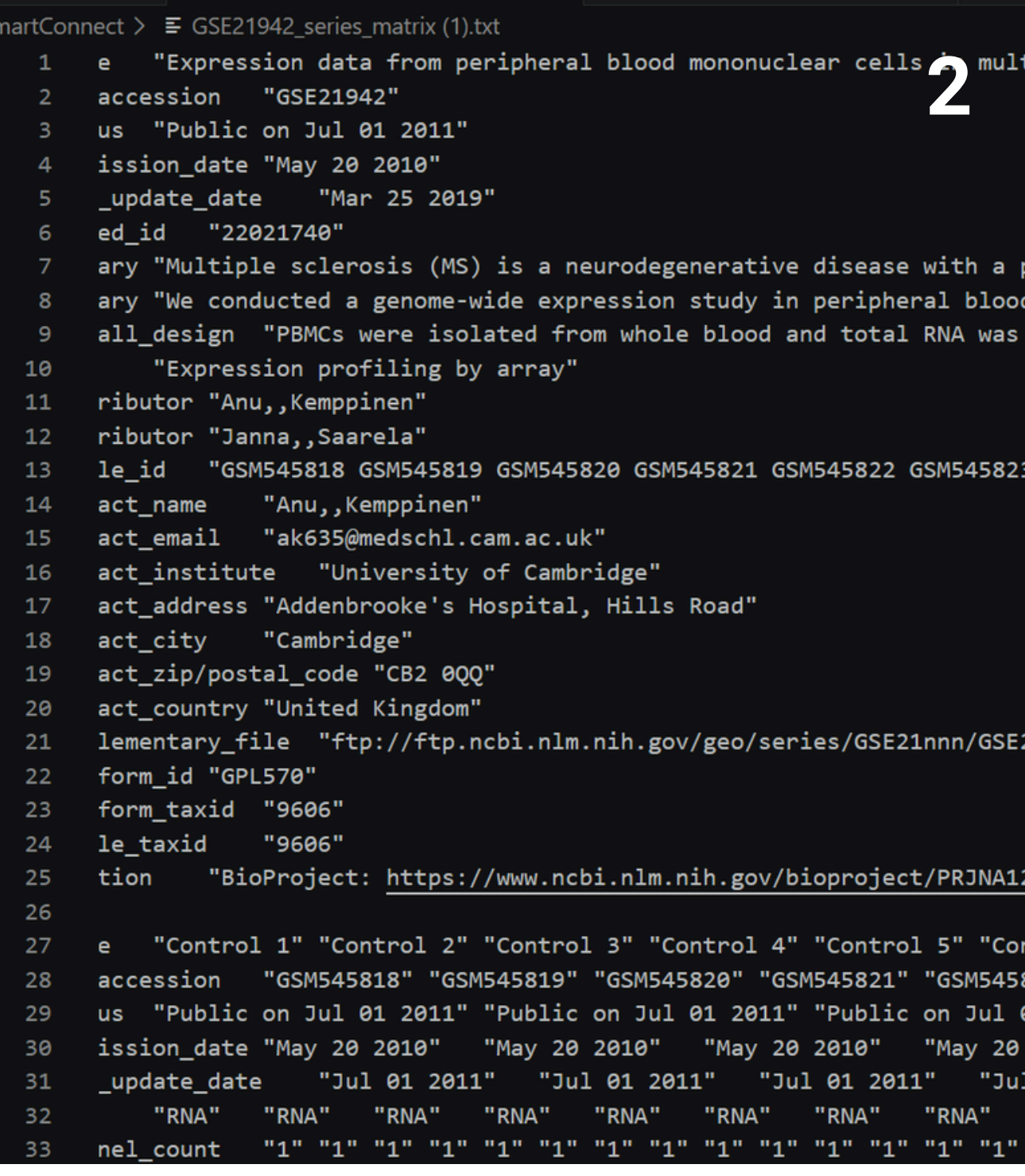
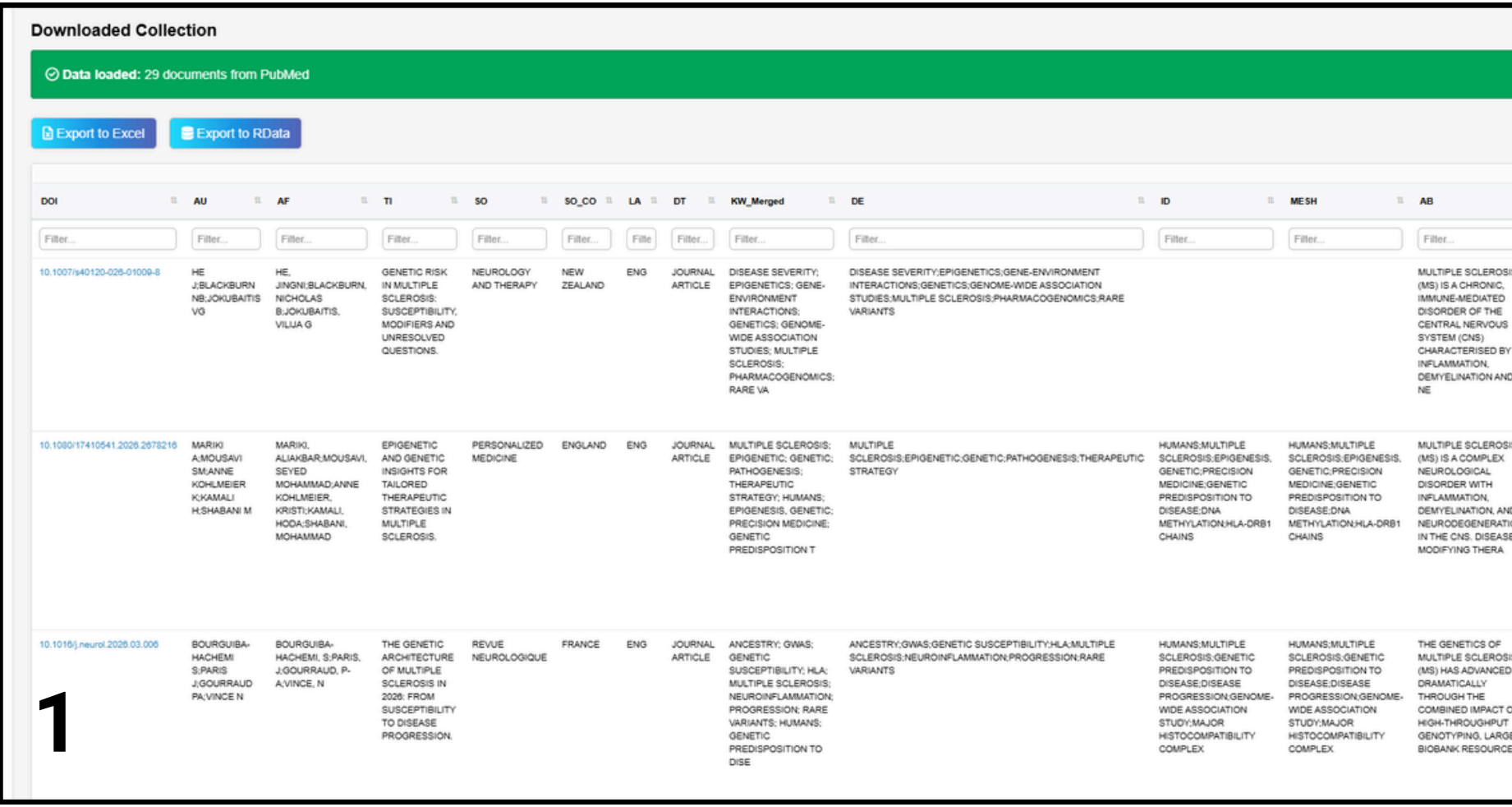
- **HLA-DRB1**: Major genetic risk factor; involved in antigen presentation (Sawcer et al., 2011).
- **IL7R**: Regulates T-cell development and survival (Gregory et al., 2007).
- **CD40**: Promotes T-cell and antigen-presenting cell communication (Blanco-Kelly et al., 2010).
- **TNFRSF1A**: Regulates inflammatory signaling and MS susceptibility (De Jager et al., 2009).

RESEARCH OBJECTIVES

- 1 Use bibliometric analysis to establish the research relevance and rationale for selecting IL7R, CD40, HLA-DRB1, and TNFRSF1A as candidate MS biomarkers.
- 2 Use differential gene expression analysis and machine learning to evaluate the suitability of these genes and identify the optimal 2-, 3-, or 4-gene combination for distinguishing MS patients from healthy controls.
- 3 Compare 2-, 3-, and 4-gene combinations using machine learning to identify the combination that most accurately distinguishes MS patients from healthy controls.

METHODOLOGY

1. **BIBLIOMETRIC ANALYSIS**
 - Using Biblioshiny, RStudio, and Biorender, we analyzed trends in recent publications of scholarly work involving MS and one of the identified genes
2. **DATA ACQUISITION**
 - Using GSE21942, a public gene expression set, we created two groups (healthy and MS). From this database (ncbi) we also downloaded the series matrix file and ran 54,696 pieces of code through visual studio to get the expression values.
3. **DIFFERENTIAL GENE EXPRESSION ANALYSIS**
 - The resulting expression values (adj.P.Val, P.Value, t, B, logF) were analyzed using differential gene expression analysis, determining patterns in upregulation/downregulation, and then visualized in Biorender
4. **MACHINE LEARNING CLASSIFICATION**
 - Using Orange, a machine learning application, we created a linear regression model, testing different combinations of genes for MS identification accuracy



RESULTS

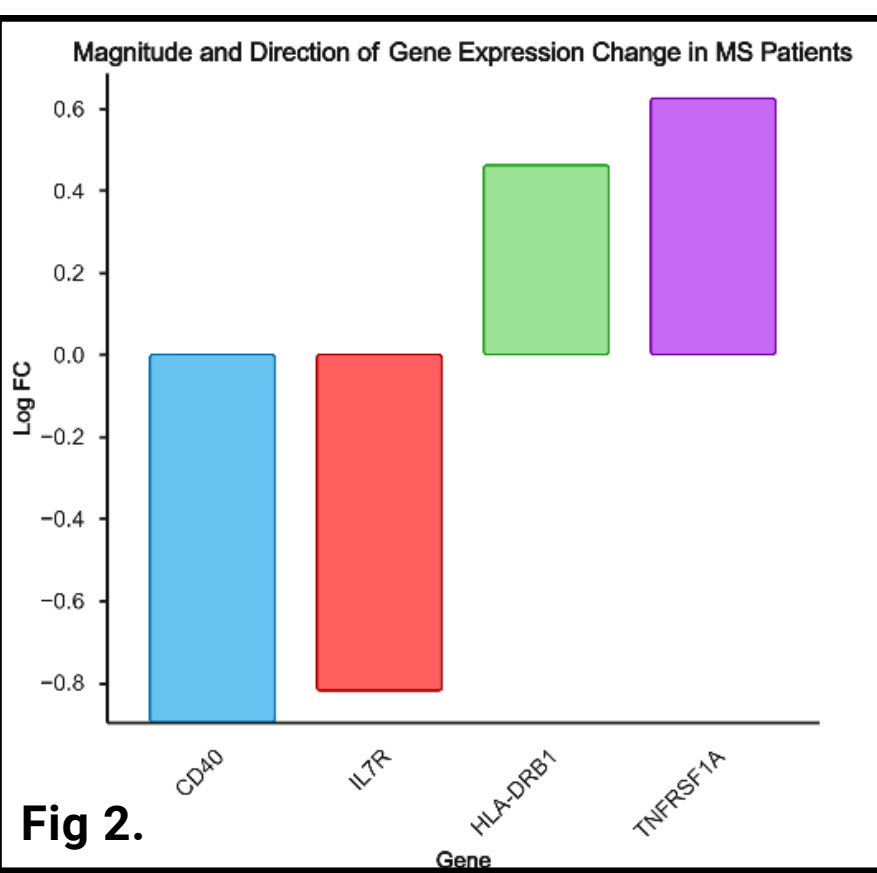
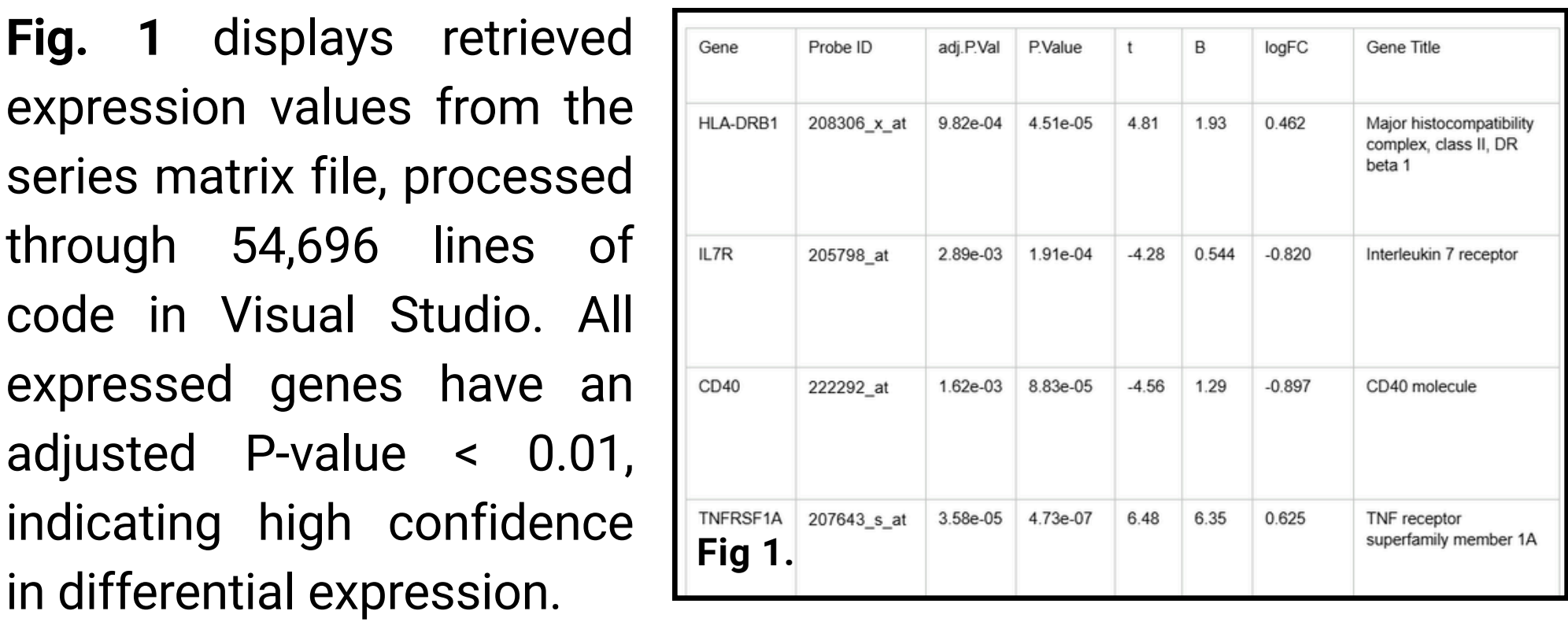


Fig. 3 demonstrates the magnitude and direction of gene-expression changes across MS patients, with logFC values ranging from -0.897 to +0.625. These values indicate small to moderate changes in gene expression from control to MS patients, revealing gene-expression changes associated with MS.

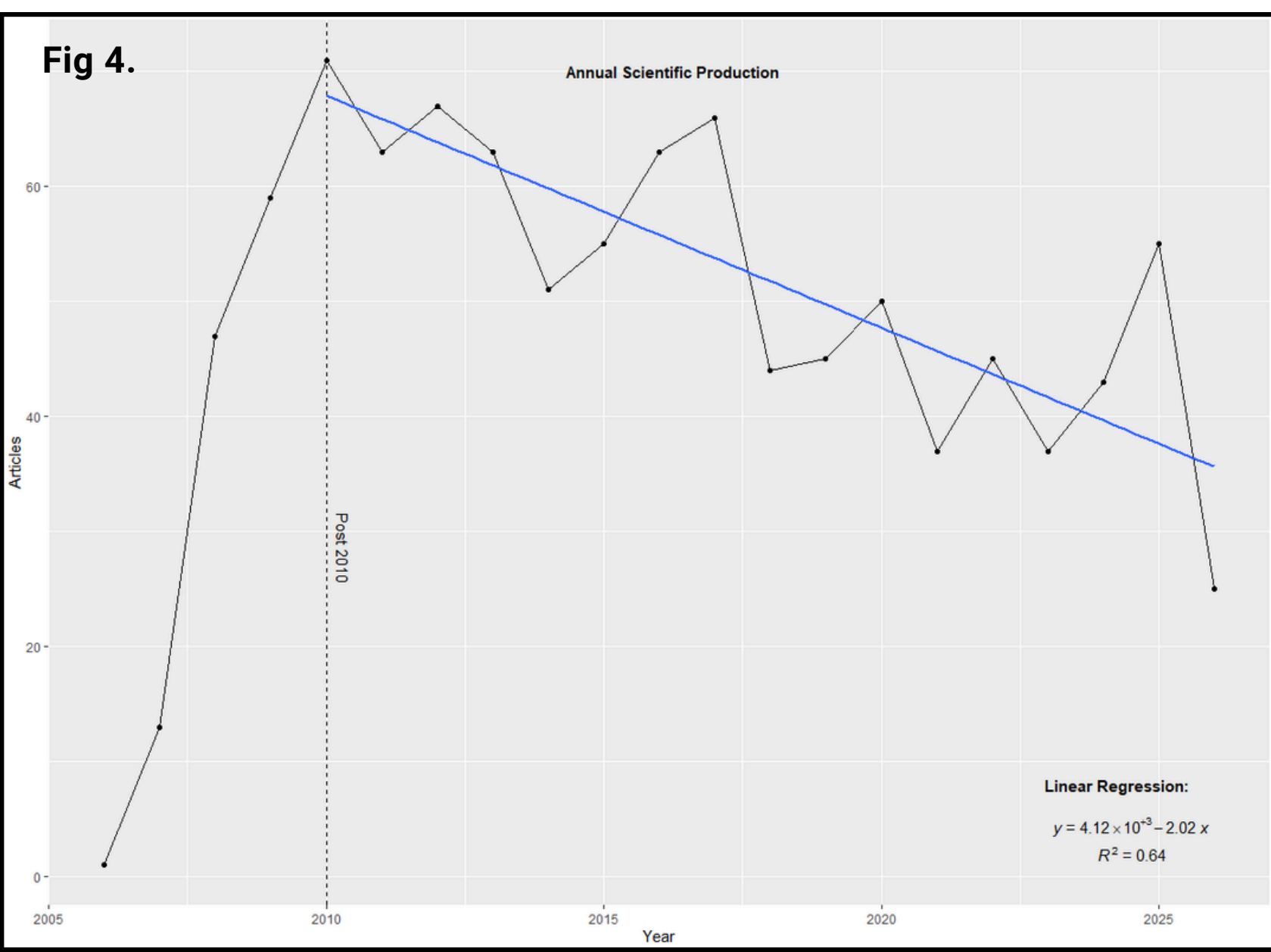
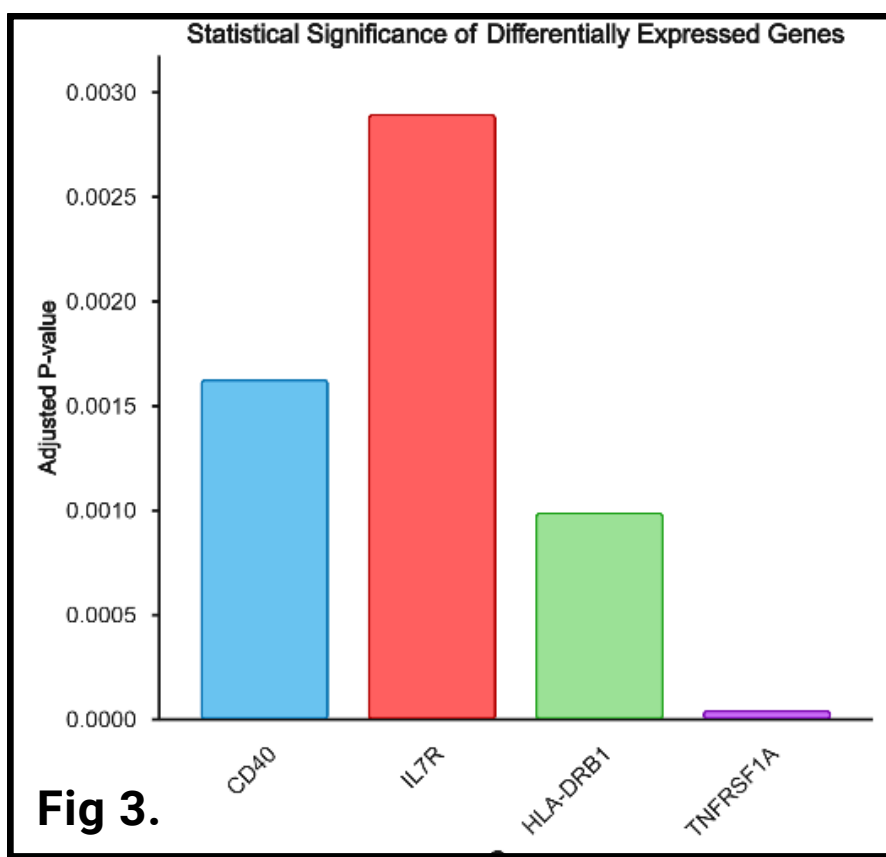


Fig. 4 demonstrates annual scientific production from 2006–2026, showing an overall increase in publications before 2010 followed by a declining trend. Post-2010 production fluctuated substantially, with a moderate negative linear trend ($R^2 = 0.64$).

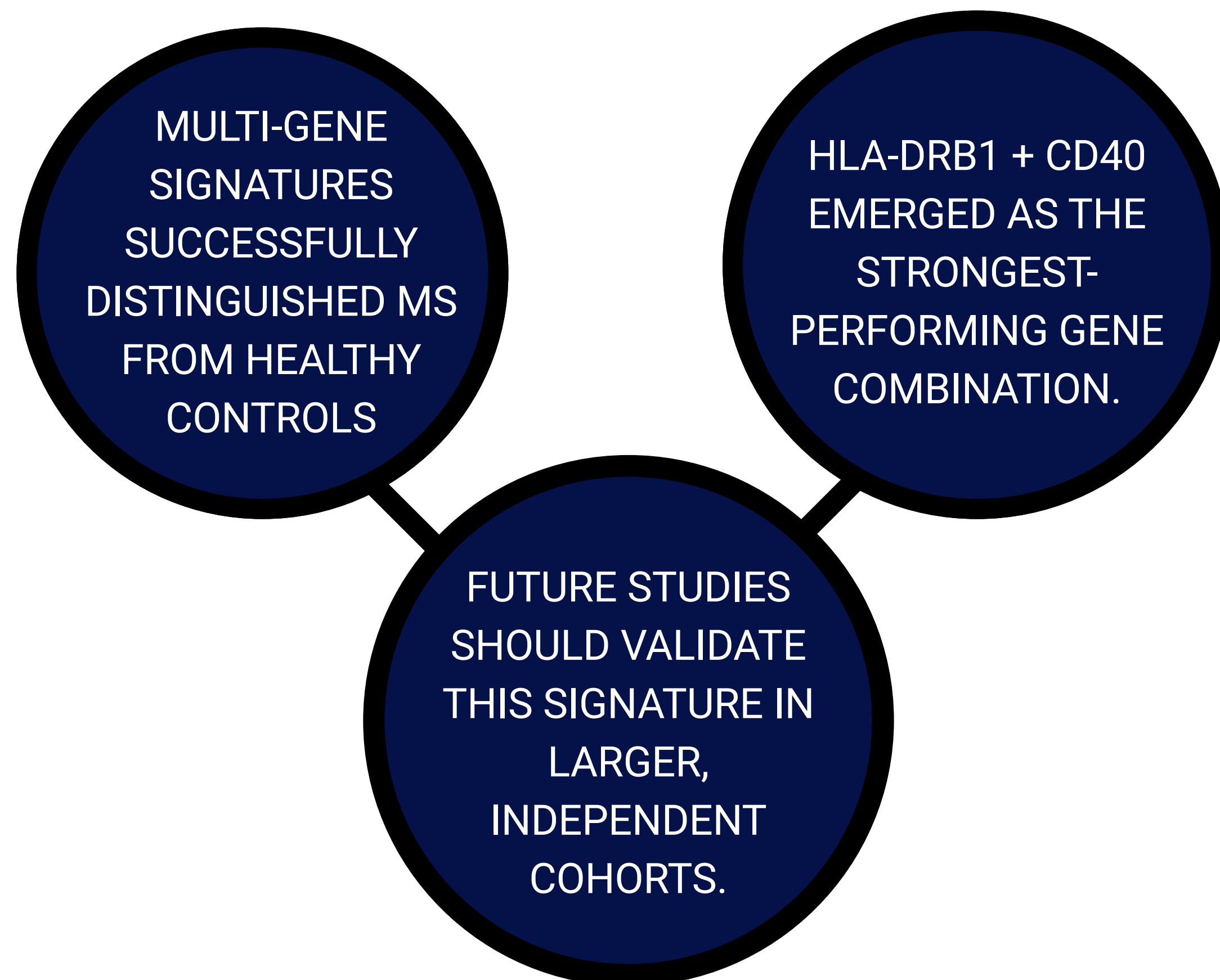
Fig 5. displays all the recorded combination outputs from the Orange linear regression model. Highlighted in yellow, the HLA-DRB1 + CD40 combination has the strongest overall performance, with the highest CA (0.966), F1 score (0.965), Precision (0.968), Recall (0.966), and MCC (0.933). Its AUC of 0.938 also indicates strong ability to distinguish between control and MS samples, making it the best-performing gene combination in the model.

Gene Combination	Genes	AUC	CA	F1	Precision	Recall	MCC
HLA-DRB1 + IL7R	2	0.850	0.828	0.828	0.830	0.828	0.857
HLA-DRB1 + CD40	2	0.938	0.966	0.965	0.968	0.966	0.933
HLA-DRB1 + IL7R + CD40	3	0.938	0.931	0.931	0.931	0.931	0.862
HLA-DRB1 + TNFRSF1A	2	0.919	0.931	0.931	0.930	0.931	0.870
IL7R + CD40	2	0.962	0.962	0.962	0.962	0.962	0.724
HLA-DRB1 + IL7R + TNFRSF1A	3	0.924	0.897	0.896	0.898	0.897	0.794
IL7R + TNFRSF1A	2	0.924	0.931	0.931	0.931	0.931	0.862
CD40 + TNFRSF1A	2	0.933	0.966	0.965	0.968	0.966	0.933
HLA-DRB1 + CD40 + TNFRSF1A	3	0.929	0.897	0.897	0.899	0.897	0.795
HLA-DRB1 + IL7R + CD40 + TNFRSF1A	4	0.905	0.897	0.896	0.898	0.897	0.794

DISCUSSION

- There is a need for further research involving MS and gene expression given the 64.79% decrease in yearly publications post-2010.
- All four genes demonstrated high confidence in differential expression (with adjusted P-values < 0.01)
- The HLA-DRB1 + CD40 combination achieved the highest diagnostic accuracy (0.966 CA and 0.938 AUC)
- The use of multiple gene-expression markers may improve MS diagnosis by relying less on a single bio-marker though future research is needed

CONCLUSION



ACKNOWLEDGMENTS & REFERENCES

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